Final Project

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I. NEWTON

a) Lame-Emden Equation

For a spherically symmetric star in Newtonian gravity, the equations of stellar structure are:

$$\frac{dm}{dr} = 4\pi r^2 d(r), \tag{1}$$

$$\frac{dP}{dr} = -\frac{Gm(r)d(r)}{r^2} \tag{2}$$

where $P(r)$ is the pressure and the $d(r)$ is the density. We also know the relationship between pressure and density is:

$$P = Kd^{1+1/n} \tag{3}$$

where K is a constant.
Beginning with eliminating the mass function to use the pressure and density relation. Multiply Eq. (2) by $r^2/d(r)$:

$$\frac{r^2}{d(r)}\frac{dP}{dr} = -Gm(r) \tag{4}$$

Differentiating with respect to r:

$$\frac{d}{dr}\left(\frac{r^2}{d(r)}\frac{dP}{dr}\right) = -G\frac{dm}{dr} \tag{5}$$

Using Eq. (1), we obtain

$$\frac{1}{r^2}\frac{d}{dr}\left(\frac{r^2}{d(r)}\frac{dP}{dr}\right) = -4\pi G d(r) \tag{6}$$

We can define the function $\theta(r)$ as

$$d(r) = \alpha\theta^n(r), \tag{7}$$

to eliminate $\frac{1}{n}$ term where α is constant. Thus, the pressure becomes

$$P(r) = K\alpha^{1+1/n}\theta^{n+1} \tag{8}$$

Taking the derivative with respect to r,

$$\frac{dP}{dr} = K(n+1)\alpha^{1+1/n}\theta^n\frac{d\theta}{dr} \tag{9}$$

Thus,

$$\frac{1}{d(r)}\frac{dP}{dr} = (n+1)K\alpha^{1/n}\frac{d\theta}{dr}. \tag{10}$$

Inserting this into Eq. (6):

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\theta}{dr}\right) = -\frac{4\pi G}{(n+1)K}\alpha^{1-1/n}\theta^n \tag{11}$$

We can simplify this equation with using scaled radius

$$r = \beta\xi, \tag{12}$$

where

$$\beta^2 = \frac{(n+1)K}{4\pi G}\alpha^{1/n-1} \tag{13}$$

Using

$$\frac{d}{dr} = \frac{1}{\beta}\frac{d}{d\xi},$$

we obtain

$$\frac{1}{\xi^2}\frac{d}{d\xi}\left(\xi^2\frac{d\theta}{d\xi}\right) + \theta^n = 0 \tag{14}$$

Thus the final result is the Lane–Emden equation of index n with boundary conditions

$$\theta(0) = 1, \qquad \left.\frac{d\theta}{d\xi}\right|_{\xi=0} = 0$$

Using the Lane–Emden change of variables,

$$\rho(r) = \rho_c\theta(\xi)^n, \qquad r = a\xi, \tag{15}$$

we see that

$$\theta = 0 \quad \Longleftrightarrow \quad \rho = 0$$

Hence the surface of the star is located at $\xi = \xi_n$, where

$$\theta(\xi_n) = 0, \tag{16}$$

and the physical radius is

$$R = a\xi_n \tag{17}$$

Using the Eq. (1)

$$\frac{dm}{dr} = 4\pi r^2 d(r) \tag{18}$$

Changing variables to ξ, using $r = \beta\xi$ and $dr = \beta\, d\xi$, we obtain

$$\frac{dm}{d\xi} = 4\pi(a\xi)^2\alpha\theta^n\, a = 4\pi a^3\rho_c\xi^2\theta^n. \tag{19}$$

Integrating from the 0 to radius ξ gives

$$m(\xi) = 4\pi\beta^3\alpha \int_0^{\xi} \xi'^2 \theta(\xi')^n \, d\xi' \tag{20}$$

We can use the Lane-Emden equation to integrate and eliminate θ^n term:

$$\xi^2\theta^n = -\frac{d}{d\xi}\left(\xi^2 \frac{d\theta}{d\xi}\right) \tag{21}$$

Substituting into the integral,

$$\int_0^{\xi} \xi'^2\theta^n \, d\xi' = -\int_0^{\xi} \frac{d}{d\xi'}\left(\xi'^2 \frac{d\theta}{d\xi'}\right) d\xi' \tag{22}$$

$$= -\left[\xi'^2 \frac{d\theta}{d\xi'}\right]_0^{\xi} \tag{23}$$

As $\theta'(0) = 0$

$$\int_0^{\xi} \xi'^2\theta^n \, d\xi' = -\xi^2\theta'(\xi) \tag{24}$$

Therefore, the enclosed mass is

$$m(\xi) = 4\pi\beta^3\alpha\left(-\xi^2\theta'(\xi)\right) \tag{25}$$

$$M = m(\xi_n) = 4\pi\beta^3\alpha\left(-\xi_n^2\theta'(\xi_n)\right) \tag{26}$$

Using $a = R/\xi_n$, we obtain

$$M = 4\pi\rho_c R^3\left(-\frac{\theta'(\xi_n)}{\xi_n}\right) \tag{27}$$

To show mass function of group of starts who shares same EOS function, we start with polytropic equation of state:

$$P = Kd(r)^{1+1/n}, \tag{28}$$

where n is fixed polytropic index and K is constant. Using (2) and (8), we can obtain:

$$\beta^2 = \frac{(n+1)K}{4\pi G}\,\alpha^{1/n-1} \tag{29}$$

$$R = \beta\xi_n \propto \beta. \tag{30}$$

Combining this with the expression for a:

$$\alpha \propto R^{\frac{2n}{1-n}} \tag{31}$$

We showed that the total mass of the star scales as

$$M \sim \alpha R^3, \tag{32}$$

Eliminating α gives the mass-radius relation:

$$M \propto R^{\frac{3-n}{1-n}} \tag{33}$$

b) M vs R of White Dwarfs

R was calculated using the equation below and the $\log(g)$ values in the datasheet.

$$g = \frac{MG}{R^2} \tag{34}$$

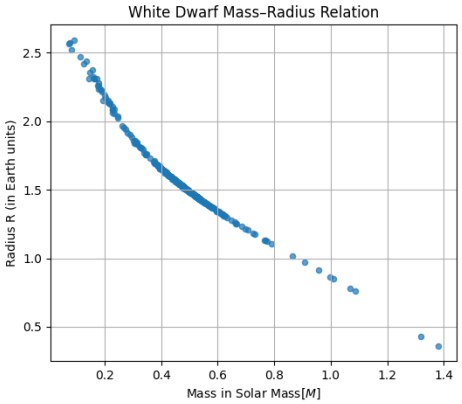


FIG. 1. Molar mass vs R

c) Mass-Radius Relation

Using Mathematica to expand the pressure for $x \ll 1$,

$$P = C\left[x\left(2x^2-3\right)\sqrt{x^2+1} + 3\sinh^{-1}(x)\right], \qquad x = \left(\frac{\rho}{D}\right)^{1/q} \tag{35}$$

and the first three terms are

$$P = \frac{8C}{5}x^5 - \frac{4C}{7}x^7 + \frac{C}{3}x^9 + O[x]^{10} \tag{36}$$

with the leading term is x^5. Thus the P is

$$P = C\left[x\left(2x^2-3\right)\sqrt{x^2+1} + 3\sinh^{-1}(x)\right], \qquad x = \left(\frac{\rho}{D}\right)^{1/q} \tag{37}$$

Using only the leading term $P \simeq \frac{8C}{5}x^5$ and the relation $x = \left(\frac{\rho}{D}\right)^{1/q}$, we obtain:

$$P = \frac{8C}{5}\left(\frac{\rho}{D}\right)^{5/q} = \frac{8C}{5D^{5/q}}\,\rho^{5/q} \tag{38}$$

Comparing with the polytropic equation of state,

$$P = K_*\rho^{1+\frac{1}{n_*}}, \tag{39}$$

We obtain

$$1 + \frac{1}{n_*} = \frac{5}{q}, \qquad K_* = \frac{8C}{5D^{5/q}} \tag{40}$$

Solving for the polytropic index,

$$n_* = \frac{q}{5-q} \tag{41}$$

For a polytropic equation of state,

$$P = K_* \rho^{1+\frac{1}{n_*}}, \tag{42}$$

Using the Eq. 27 and $R = a\,\xi_1$ where ξ_1 is the first zero of the Lane–Emden function, we obtain:

$$a = \left[\frac{(n_*+1)K_*}{4\pi G}\right]^{1/2} \rho_c^{\frac{1-n_*}{2n_*}} \tag{43}$$

Eliminating the central density ρ_c from the above expressions gives a power law relation between mass and radius:

$$R \propto K_*^{\frac{n_*}{3-n_*}} M^{\frac{1-n_*}{3-n_*}} \tag{44}$$

Taking the logarithm, we obtain

$$\log R = \frac{1-n_*}{3-n_*} \log M + \text{const.} \tag{45}$$

Thus, the slope b of a $\log R$ versus $\log M$ plot is

$$b = \frac{1-n_*}{3-n_*} \tag{46}$$

Solving for the polytropic index n_* gives

$$n_* = \frac{1-3b}{1-b} \tag{47}$$

Therefore, by fitting a power law to the portion of the observed white dwarf mass-radius relation,

$$R \propto M^{\,b}, \tag{48}$$

we can infer the effective polytropic index n_*. Using numerical methods, I obtained b=-0.3108, which gives n=1.4742 and q=2.97911. When we round q to the nearest integer q=3. Fitting the curve for low mass,

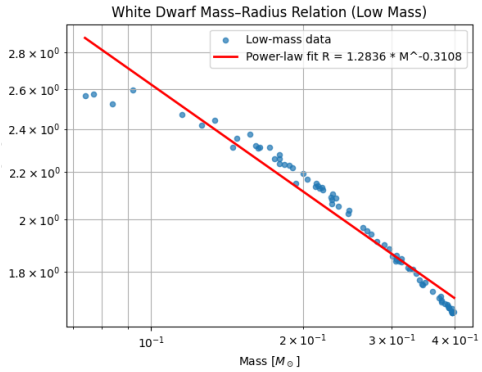


FIG. 2. M vs R for low M

d) Interpolation for M vs R

Using the part c, we can obtain C:

$$C = \frac{5KD^5}{8q}, \tag{49}$$

leaving a single free parameter D. We have to determine D which fits the white dwarf mass and radius data.

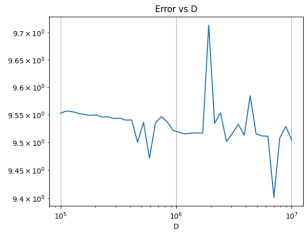


FIG. 3. Error vs D

From the low-mass white dwarf fit, we obtain

$$n = 1.474, \qquad q = 2.98.$$

Minimizing the total fitting error in Part (d) yields a best-fit value of the EOS parameter

$$D_{\text{best}} = 7.90 \times 10^7.$$

The theoretical expectation is

$$D_{\text{theory}} = 7.01 \times 10^7.$$

$$C_{\text{theory}} = 6.02 \times 10^{22}, \qquad C_{\text{best}} = 1.49 \times 10^{23}.$$

II. EINSTEIN

a) M vs R

We use the parametric polytropic equation of state:

$$P = K\rho_r^{\Gamma}, \qquad \rho = \rho_r + \frac{K}{\Gamma-1}\rho_r^{\Gamma}, \tag{50}$$

where P is the pressure, ρ is the total energy density, $\Gamma = 1.3569$ is fixed, and K is a constant. The constant K is fixed by a reference provided in the problem:

$$P_0/c^2 = 1.5689{\times}10^{31}\ \text{dyn/cm}^2 \quad \text{when} \quad \rho_{r,0} = 10^{13}\ \text{g/cm}^3. \tag{51}$$

Since $P_0 = K\rho_{r,0}^{\Gamma}$, we obtain

$$K = \frac{P_0}{\rho_{r,0}^{\Gamma}}. \tag{52}$$

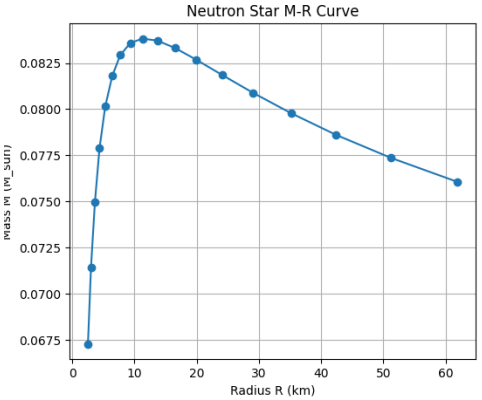
Then we integrate TOV equations:

$$\frac{dm}{dr} = 4\pi r^2 \rho, \tag{53}$$

$$\frac{d\nu}{dr} = \frac{2m + 4\pi r^3 p}{r(r-2m)}, \tag{54}$$

$$\frac{dp}{dr} = -\frac{m + 4\pi r^3 p}{r(r-2m)}(\rho + p) = -\frac{1}{2}(\rho + p)\frac{d\nu}{dr}. \tag{55}$$

outward from the center of the star with the initial conditions given as $m(0) = 0, p(0) = p_c, \nu(0) = 0$



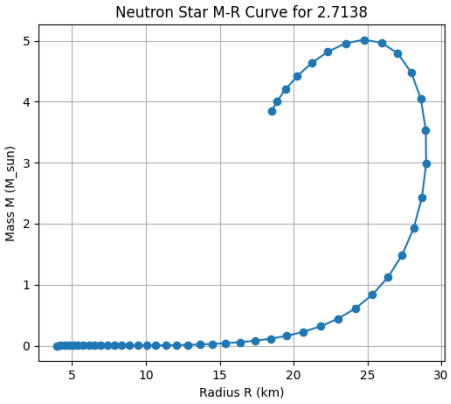


FIG. 4. M vs R for neutron star

b Baryonic Mass

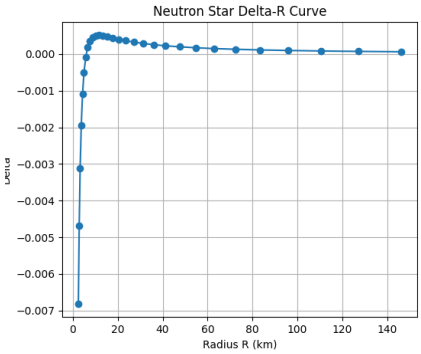
Baryonic Mass and Binding Energy

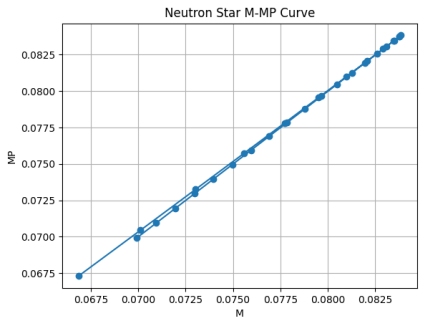
The total rest mass of the particles alone, referred to as the baryonic mass M_P, is obtained by integrating

$$m_P'(r) = 4\pi r^2 \rho_r \left(1 - \frac{2m(r)}{r}\right)^{-1/2}, \tag{56}$$

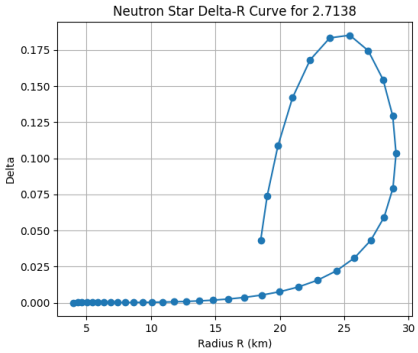
where ρ_r is the rest-mass density. The fractional binding energy is defined as

$$\Delta \equiv \frac{M_P - M}{M}, \tag{57}$$





which measures the fraction of the total mass. The appearance of the cusp indicates that both the gravitational mass and the baryonic mass reach their maximum values, beyond which no stable equilibrium solutions exist.



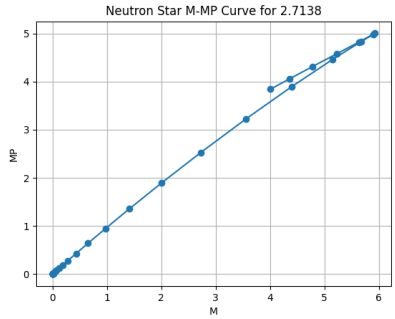


FIG. 5. Δ vs R and M vs M_p for neutron star

c)

Stability of Neutron Stars

Stability criterion for neutron star is

$$\frac{dM}{d\rho_c} > 0 \quad \text{(stable)}, \qquad \frac{dM}{d\rho_c} < 0 \quad \text{(unstable)}. \tag{58}$$

The maximum neutron star mass allowed by the chosen equation of state is given by the peak of the $M(\rho_c)$ curve, where $dM/d\rho_c = 0$.

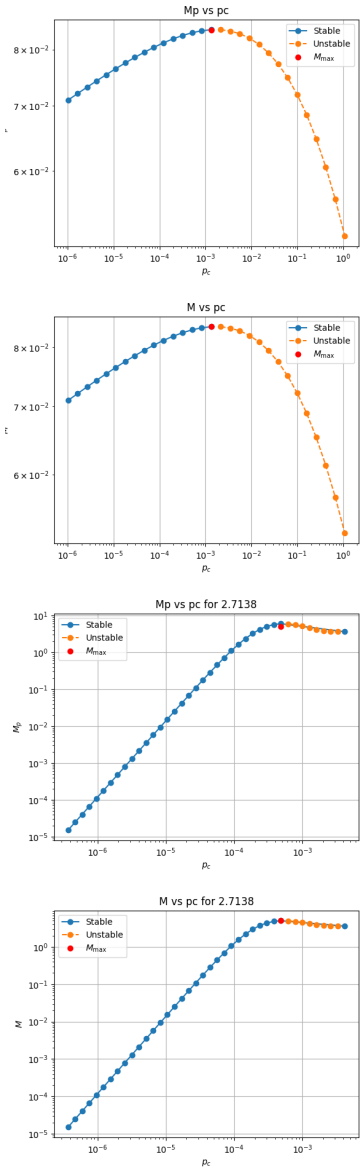


FIG. 6. M_p vs ρ_c and M_p vs ρ_c for neutron star

Numerical results give for $\Gamma = 1.3569$, the maximum mass occurs at a central density

$$\rho_{c,\max} = 1.33 \times 10^{-3}, \tag{59}$$

with corresponding gravitational and baryonic masses

$$M_{\max} = 0.08380, \qquad M_{P,\max} = 0.08384. \tag{60}$$

For $\Gamma = 2.7138$, the maximum mass is reached at

$$\rho_{c,\max} = 4.856 \times 10^{-4}, \tag{61}$$

with

$$M_{\max} = 5.0049, \qquad M_{P,\max} = 5.9320. \tag{62}$$

These values explains cusps in part b, where no solution exists after some M values. We can see that values in both graphs correlate.

d) K vs M_{max}

Allowed K values are values after 1.5 approximately for gamma 1.3569.

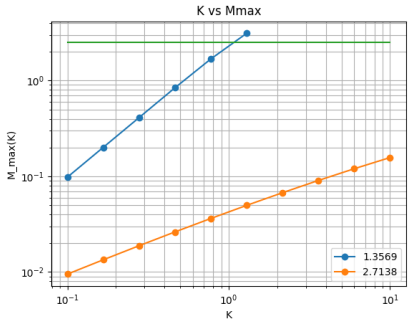


FIG. 7. K vs Mmax

e)

Using Mathematica to obtain ν using the equation:

$$\frac{d\nu}{dr} = \frac{2M}{r(r-2M)}. \tag{20}$$

We integrate this equation from R to an arbitrary radius $r > R$:

$$\nu(r) - \nu(R) = \int_R^r \frac{2M}{r'(r'-2M)}\, dr'. \tag{63}$$

and the integral is

$$\int_R^r \left(\frac{1}{r'-2M} - \frac{1}{r'}\right) dr' = \left[\ln(r'-2M) - \ln r'\right]_R^r. \tag{64}$$

Evaluating the limits and adding $\nu(R)$ gives

$$\nu(r) = \ln\left(\frac{r-2M}{r}\right) - \ln\left(\frac{R-2M}{R}\right) + \nu(R). \tag{65}$$

Finally, simplifying,

$$\nu(r > R) = \ln\left(1 - \frac{2M}{r}\right) - \ln\left(1 - \frac{2M}{R}\right) + \nu(R) \tag{66}$$